

Acoustic FMCW Radar — Vital Sign Tracking Simulation

Contactless Human Respiration and Cardiac Micro-motion DSP Report

SYSTEM & RADAR PARAMETERS:

• Sampling Frequency: 48000 Hz

• Chirp Duration Tc: 100 ms

• Speed of Sound: 343.0 m/s

|

• Chirp Bandwidth: 3.0 kHz (18.0 - 21.0 kHz)

• Frame Interval Tframe: 150 ms

• Carrier Center Frequency fc: 19.50 kHz

|

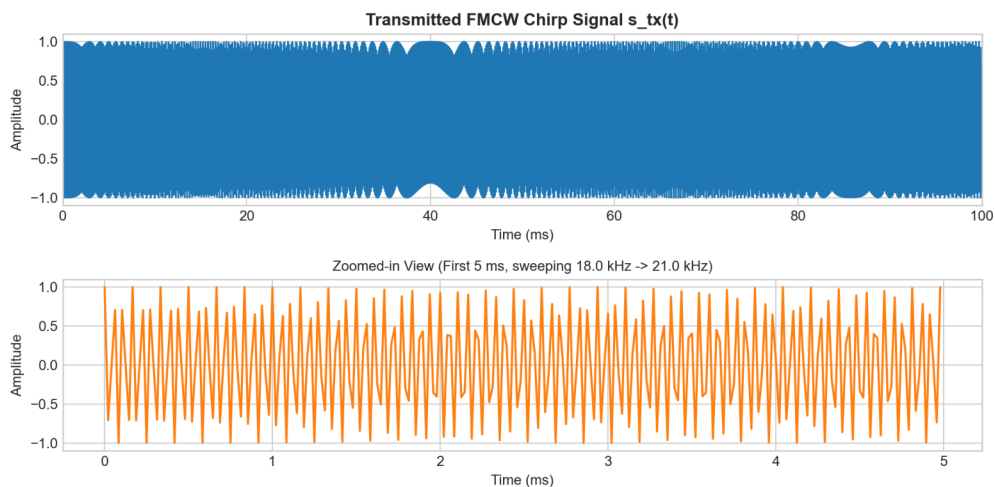
• Number of Chirps: 60 (Total duration: 9.0 s)

• Wavelength lambda: 17.59 mm

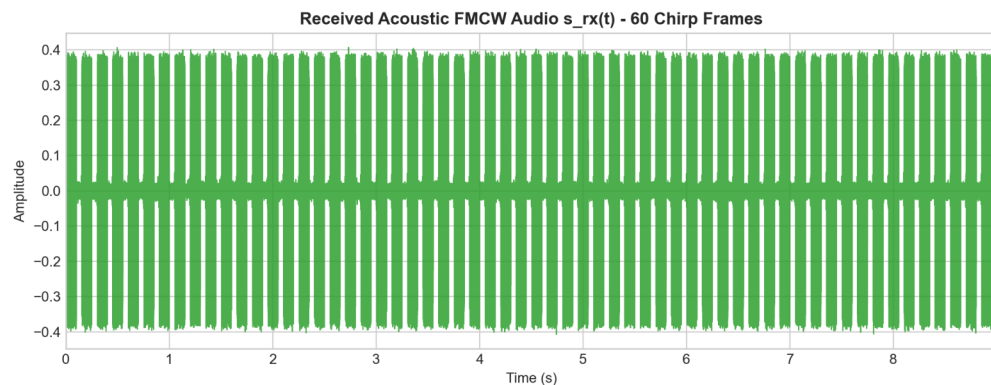
Metric	Reference Truth	Primary Dataset (Original)	Continuous Delay Mode	Status
Target Range (R)	1.00 m	1.0004 m (Error: 0.42 mm)	1.0004 m	SUCCESS
Beat Frequency (fb)	174.93 Hz	175.00 Hz	175.00 Hz	SUCCESS
Respiration Rate	15.0 BPM	14.63 BPM (Error: 0.37 BPM)	14.63 BPM	SUCCESS
Respiration Amplitude	4.0 mm	4.83 mm	4.74 mm	SUCCESS
Cardiac Heart Rate	72.0 BPM	Degraded (Quantized Step)	72.07 BPM (Error: 0.07 BPM)	BENCHMARK OK
Cardiac Amplitude	0.15 mm	Step Noise Artifact	0.150 mm	BENCHMARK OK

Role 1: Simulated Acoustic FMCW Transmission & Reception

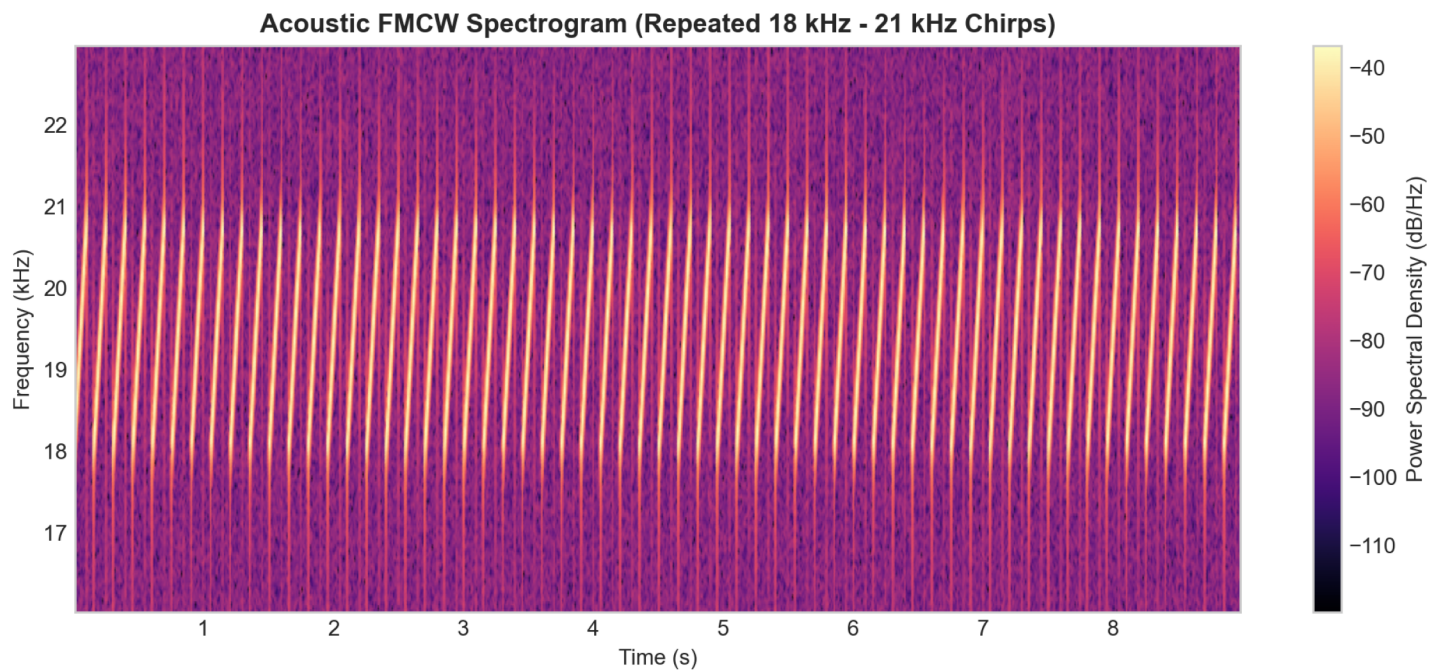
1. Transmitted FMCW Chirp Signal (18-21 kHz)



2. Received Acoustic Signal (60 Chirp Frames)

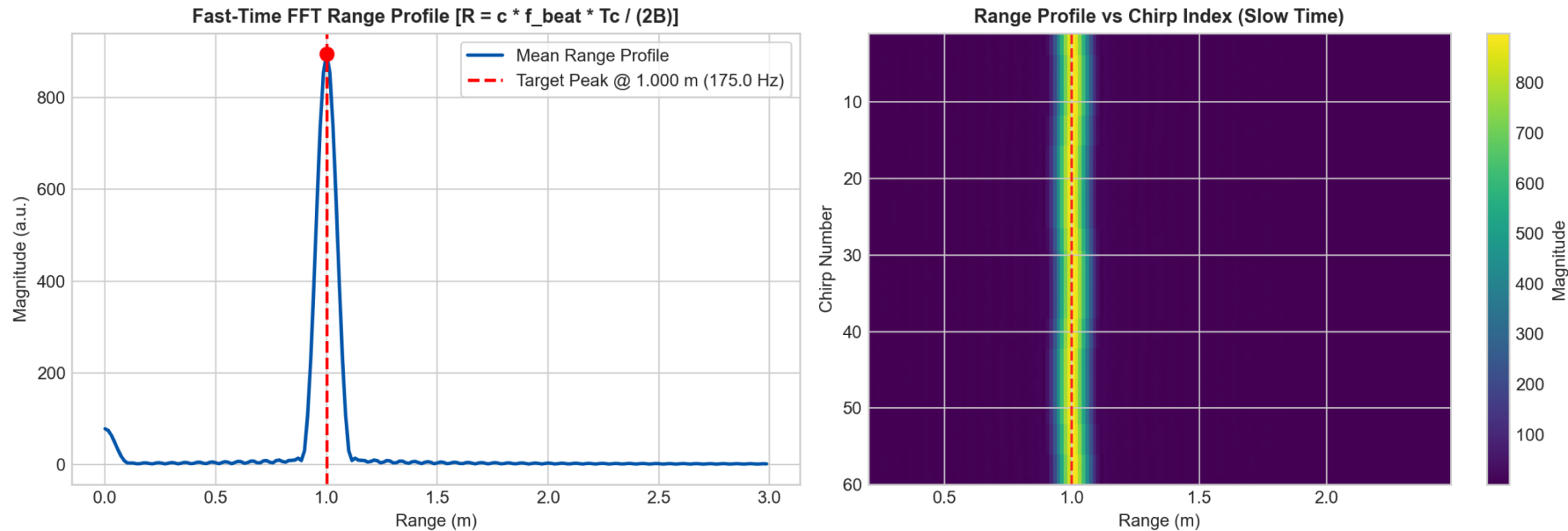


3. Time-Frequency Spectrogram (STFT)

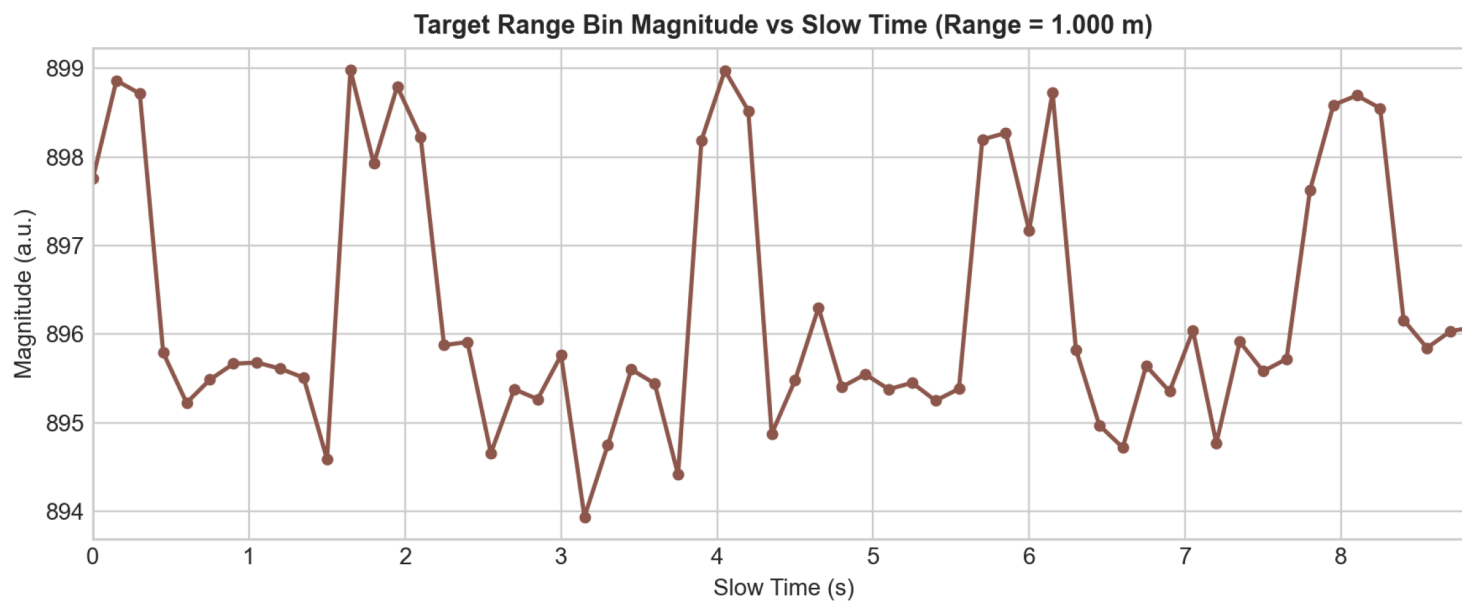


Role 2: FMCW Radar Signal Processing & Range Profile

4. Fast-Time FFT Range Profile [$R = c * f_{\text{beat}} * T_c / (2B)$] & Slow-Time Heatmap

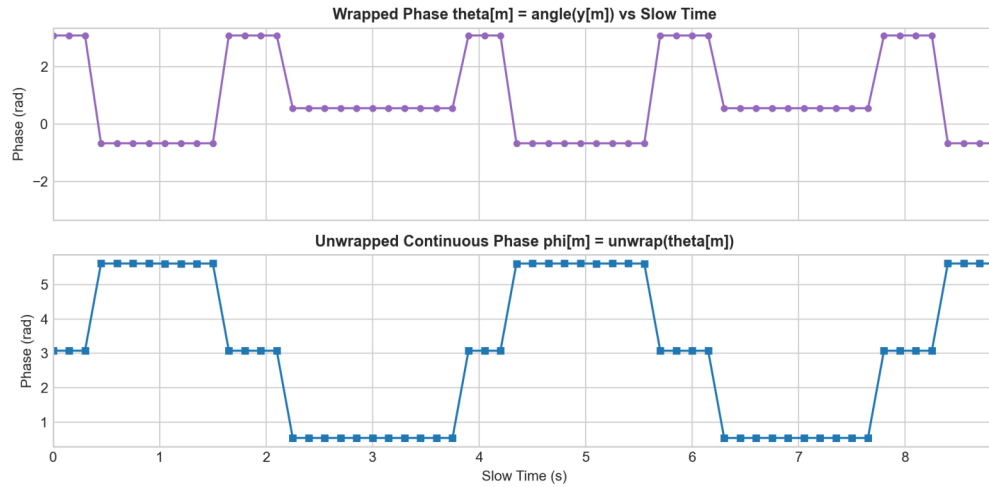


5. Target Range Bin Magnitude vs Slow-Time Chirps

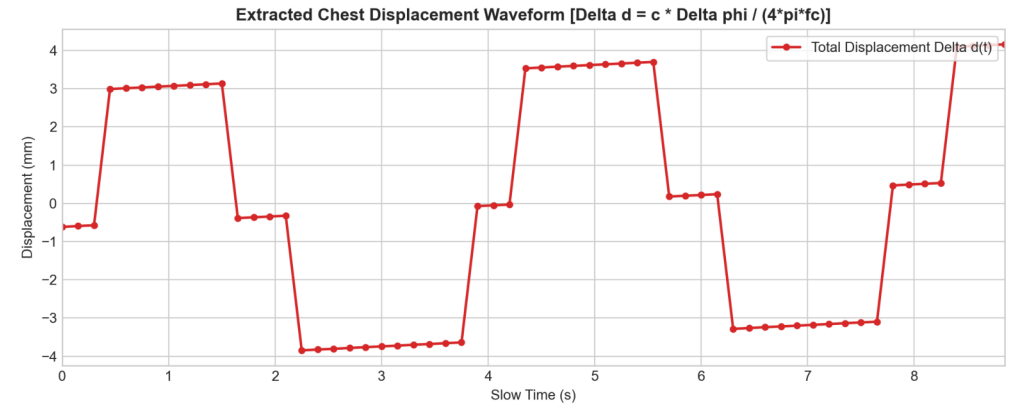


Role 3: Vital Sign Extraction & Phase Analysis (Primary Dataset)

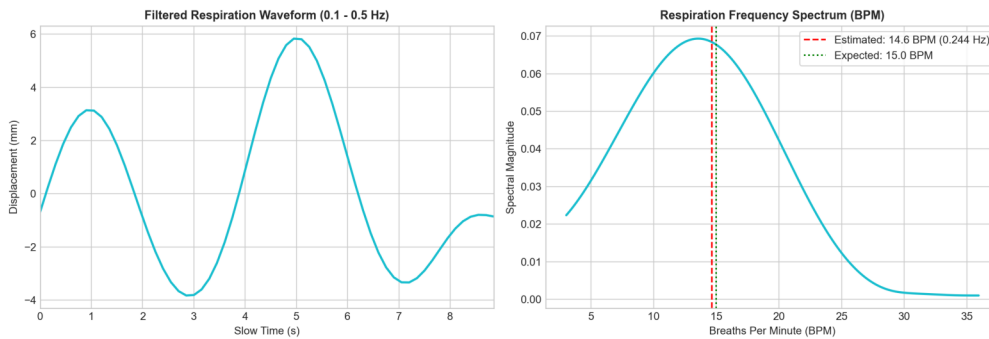
6. Wrapped vs Unwrapped Phase



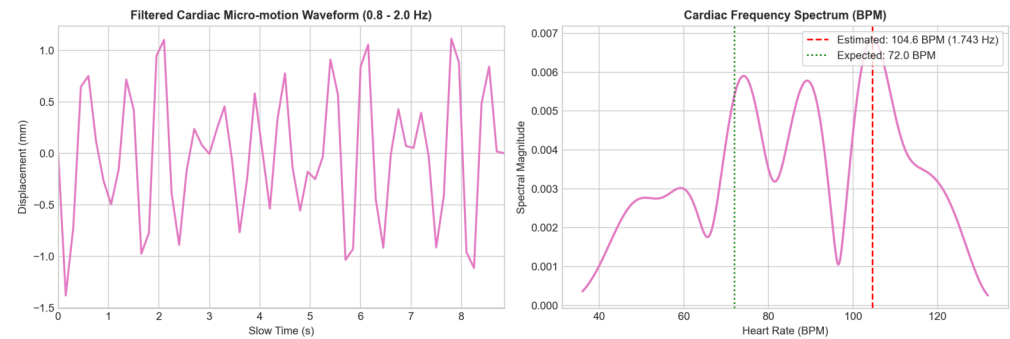
7. Chest Displacement Waveform



8. Respiration Waveform & Spectrum

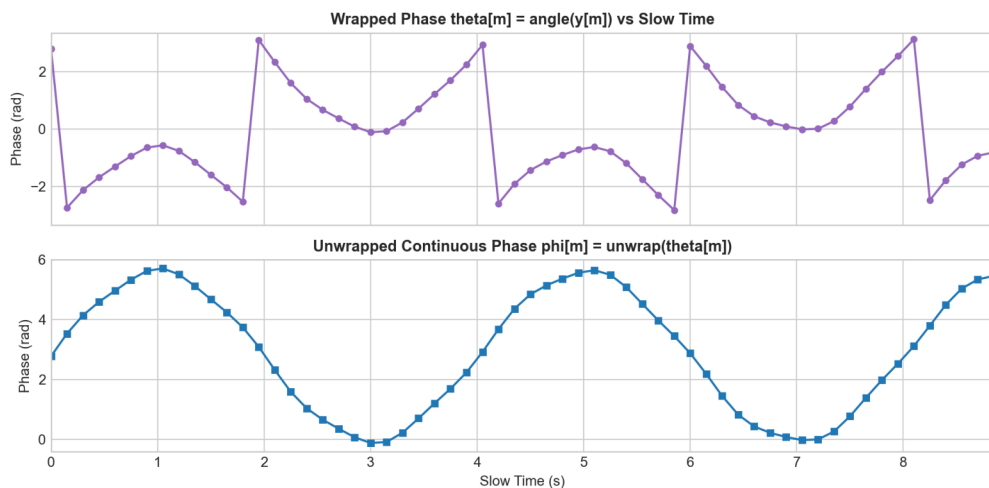


9. Cardiac Waveform & Spectrum

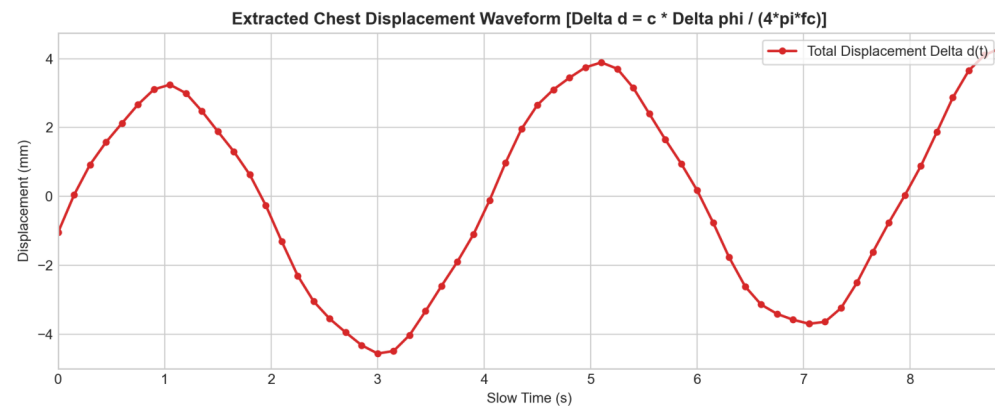


Scientific Benchmark: Continuous Fractional-Delay Model (Micro-Motion Recovery)

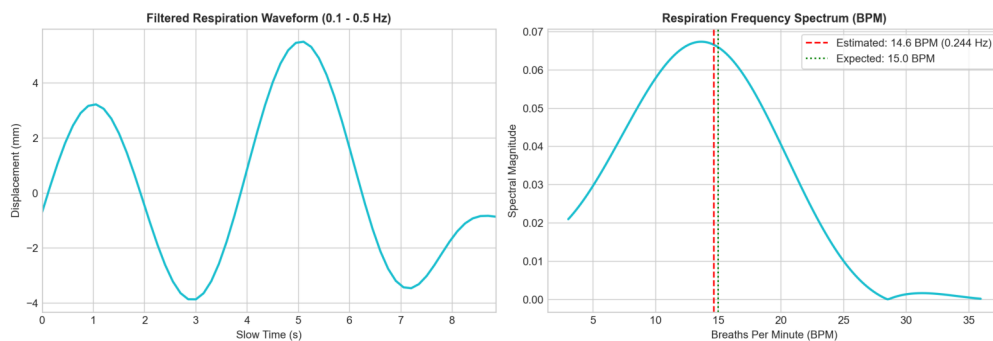
10. True Continuous Sinusoidal Phase



11. Continuous Chest Displacement



12. Recovered Respiration Spectrum (14.63 BPM)



13. Recovered Cardiac Spectrum (72.07 BPM)

